160	5500	0.698	17900	1.160
167	5720	0.681	19000	1.231
177	5930	0.665	20800	1.348
177	6350	0.635	19900	1.29
177	6570	0.620	19400	1.257
177	6800	0.603	18900	1.225

Fig 7.3

[B1 for correct answer]

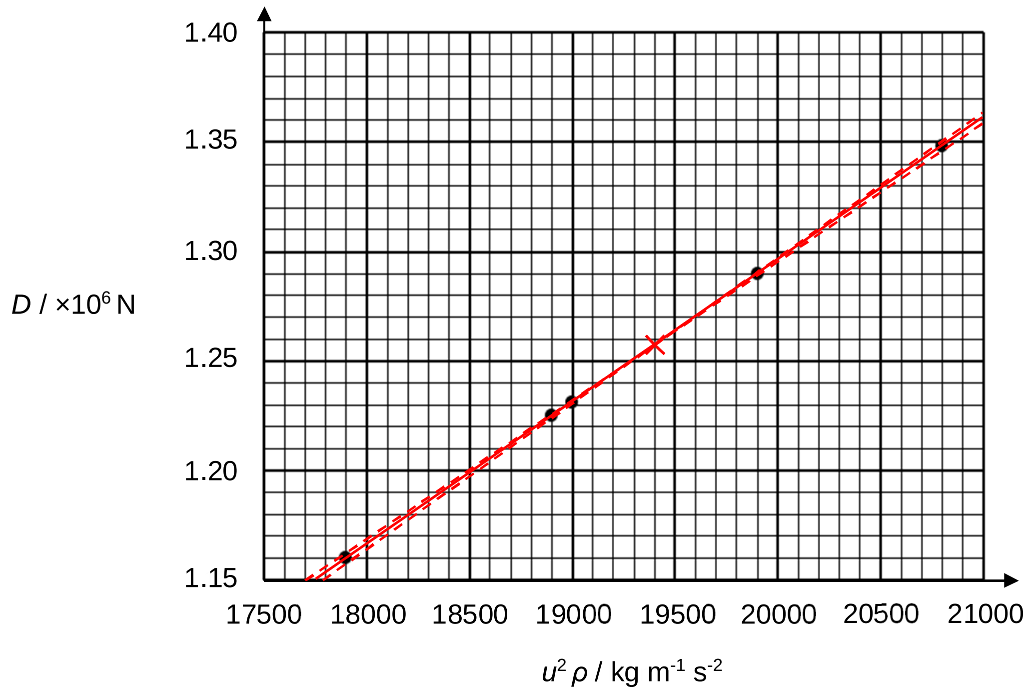

Fig 7.4

(iii)

$$\frac{1}{2}C_D S = \frac{y_2 - y_1}{x_2 - x_1} \qquad \text{[C1: relating } \frac{1}{2}C_D S \text{ to gradient]}$$

$$C_D = \frac{2}{S}\frac{(y_2 - y_1)}{(x_2 - x_1)}$$

Readoffs $(17900, 1.16)$ and $(20800, 1.35)$

$$C_D = \frac{2}{162}\frac{(1.35 - 1.16)\times 10^6}{(20800 - 17900)}$$

$$= \frac{2}{162}(65.517)$$

$= 0.809$ (no units) [A1: within 0.785 to 0.825]

(iv) Quantitative calculation to show that the vertical intercept is close to zero. [B1]
Deduce that the expression is correct with the straight line graph. [B1]
OR
Using two sets of values read from the line of best fit, it can be shown that the ratio of *D* and $u^2 \rho$ is approximately constant [B1] (calculations must be presented)
and the expression shows that they are directly proportional to each other [B1]